



First Semester 2026-27

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : CE G534

Course Title : Pavement Material Characterization

Instructor-in-charge : Dr. Vijay Bajirao Kakade

[Team of Instructors] :-

1. Course description: [as per the Bulletin]

This course covers topics related to characterization of pavement materials including soil, aggregates, bitumen, bituminous mixtures and pavement quality concrete (PQC). Soil characterization includes description of characteristics and desirable properties, classification, compaction, strength and modulus of subgrade soil. For aggregates, origin and production, mineralogy, physical properties and aggregate properties for mix design would be discussed. Detailed description of bituminous binder would include origin, production and its application, chemical composition, physical and rheological properties, grading system, modification of bitumen, cutbacks and emulsions. Bituminous mixtures include production, desirable properties, mix design and performance testing. For PQC, materials and selection of mix proportions would be covered along with design example.

2. Scope and Objective:

The main objective of this course is to understand the different material components typical in pavement and learn about the importance of the properties of these components for effective functioning of the pavement. In this course, we would also look into the guidelines outlined for testing, specifications and characterization of various materials used in pavements (both nationally and internationally). A brief about the ongoing worldwide research on various material aspects of pavements will also be covered as a part of different lectures.

3. Course learning outcomes:

- a. . To characterize pavement materials
- b. . To perform bituminous mix design
- c. . To perform mix design of concrete

4. Text books and references:

TB1. Atkins, N. Harold, *Highway Materials, Soils and Concretes*, Fourth Edition, 2002, Prentice-



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TB2. Yoder E. J. and M. W. Witzczak, Principles of Pavement Design, 2nd Edition, 1975, John Wiley & Sons, Inc

R1 : Huang, Y.H., *Pavement Analysis and Design*, Pearson, 2004

R2 : Khanna S. K., Justo, C.E.G and Veeraragavan, A., *Highway Material Testing Laboratory Manual*, Nem Chand & Bros.

R3 : Chakroborty, P. and Das, A., *Principles of Transportation Engineering*, PHI, New Delhi.

R4 : Punmia, B.C., Jain, A.K., Jain A.K. (2017). *Soil Mechanics and Foundation*. Laxmi Publications Pvt. Ltd.

R5 : B.M. Das, K. Sobhan (2016). *Principles of Geotechnical Engineering, Ninth Edition*. Cengage Learning.

R6 : Hunter, R. N., Self, A., Read, J., & Hobson, E. (2015). *The Shell Bitumen Handbook*. London, UK, Ice Publishing.

R7 : Robert, F.L., Kandhal P.S., Brown, E.R., Lee D.Y., Kennedy, T.W., Hot Mix Asphalt Materials, Mixture, Design and Construction, NCAT

R8 : Kandhal PS, Bituminous Road Construction in India, PHI, 2016

R9 : Shetty M.S., Concrete Technology – Theory and Practice, S.Chand.

R10 : Relevant Specifications and Codes

5. Method of conduct of the course: Power point presentation

6. Course Plan:

Week no.	Module	Lect. no.	Reference	Learning Outcome
1.	Soil – Desirable properties, classification, compaction, strength and modulus	1-3	T1, T2, R1, R2, R3, R4, R5, R10	Characterization of subgrade soil and assessment of strength properties. Used as a design input in pavement design.
2.	Aggregates – Geological sources & rock types, production and storage, sampling,	4-6	T1, R2, R7, R10	Effect of source and production process on aggregate properties and the applicability of the selected aggregates for production of bituminous mixtures



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3.	Aggregates – mineralogy, physical properties, properties for mix design and specification limits	7-9	T1, R2, R7, R10	Effect of source and production process on aggregate properties and the applicability of the selected aggregates for production of bituminous mixtures
4.	Bitumen – Origin and uses, production and storage, chemical composition	10-12	T1, T2, R1, R2, R3, R6, R7, R8, R10	Understanding the origin and production process of bitumen.
5.	Bitumen – physical and rheological properties,	13-15	T1, T2, R1, R2, R3, R6, R7, R8, R10	Characterizing the bitumen based on its fundamental properties and assigning different grades for selection of bitumen based on project-specific conditions
6.	Bitumen – Grading System	16-18	T1, T2, R1, R2, R3, R6, R7, R8, R10	Characterizing the bitumen based on its fundamental properties and assigning different grades for selection of bitumen based on project-specific conditions
7.	Bitumen – polymer modified bitumen	19-21	T1, T2, R1, R2, R3, R6, R7, R8, R10	Characterizing the bitumen based on its fundamental properties and assigning different grades for selection of bitumen based on project-specific conditions
8.	Bitumen – cutbacks and emulsions	22-24	T1, T2, R1, R2, R3, R6, R7, R8, R10	Characterizing the bitumen based on its fundamental properties and assigning different grades for selection of bitumen based on project-specific conditions
9.	Bituminous Mixtures – Production	25-27	T1, T2, R1, R2, R3, R6, R7, R8, R10	Assigning a job mix formula for the selected aggregate, aggregate gradation and bitumen. Testing of the resistance of bituminous mixtures to commonly observed distresses.
10.	Bituminous Mixtures – desirable properties	28-30	T1, T2, R1, R2, R3, R6, R7, R8, R10	Assigning a job mix formula for the selected aggregate, aggregate gradation and bitumen. Testing of the resistance of bituminous



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				mixtures to commonly observed distresses.
11.	Bituminous Mixtures – volumetric properties, mix design (Marshall and Superpave)	31-33	T1, T2, R1, R2, R3, R6, R7, R8, R10	Assigning a job mix formula for the selected aggregate, aggregate gradation and bitumen. Testing of the resistance of bituminous mixtures to commonly observed distresses.
12.	Bituminous Mixtures – performance testing and field verification	34-36	T1, T2, R1, R2, R3, R6, R7, R8, R10	Assigning a job mix formula for the selected aggregate, aggregate gradation and bitumen. Testing of the resistance of bituminous mixtures to commonly observed distresses.
13.	Pavement Quality Concrete – Materials	37-39	T1, T2, R1, R9, R10	Selection of materials and its respective proportions for the mix design of PQC.
14.	Pavement Quality Concrete – selection of mix proportions	40-42	T1, T2, R1, R9, R10	Selection of materials and its respective proportions for the mix design of PQC.

7. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Quiz	21%	30 min	Quiz 1: 3/9/26 Quiz 2: 1/10/26 Quiz 3: 19/11/26	All quizzes are mandatory. Closed book
Laboratory experiment	19%	-	TBA	Open book
Mid Semester	25%	90 min	08/10/2026	Open book
Comprehensive	35%	180 min	12/12/2026	Closed book

Criterion for NC: If the student absents herself/ himself even in one of the evaluative components (listed in the evaluation scheme) entirely, his / her performance shall be reported as NC (Not Cleared) regardless of his/



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her final total score in the course (See Clause 4.19 of BITS Academic Regulations). Students gaining less than 10% overall, may be awarded NC grade.

8. Chamber consultation hours: Tuesday 11 am – 12 pm

9. Makeup Policy:

- Make up will be granted only on genuine reasons and will be considered on a case by case basis. However, prior permission is a must. It should follow the official request procedure. No other mod of request shall be entertained.
- For medical cases, a certificate from the concerned physician from the Medical Centre must be produced. (Subjected to Academic regulations)
- Make up policy is only applicable to Mid – Semester test and Comprehensive Examination. It is not applicable to tests, assignment and term paper.

Dr. Vijay Kakade

Instructor-in-charge

Course No. CE G534